AVR

Embedded Systems based on AVR Diploma [AVR Diploma]

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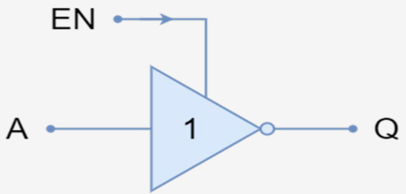
Digital Input Output

A Digital Input Output is a peripheral that deals with digital signals, either by generating a digital signal (Output Mode) or by receiving it (Input Mode).
The basic block unit for the DIO pin is the Tri-State Buffer. Any DIO pin is consisting of a Tri-State buffer as a main component. The Tri-State buffer controls the direction of the data, A to B or B to A.

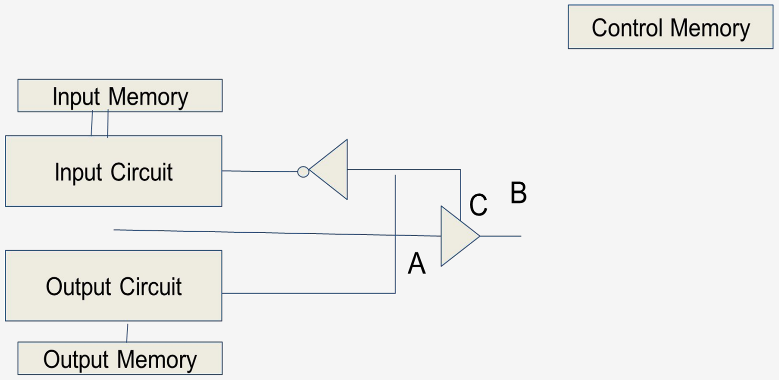


EN	Output
1	A Q
0	A Q

DIO Block Diagram

Writing 0 To C, Enables the Input Circuit and make the Buffer direction to (A <- B) which makes the PIN in Input Mode.

Writing 1 To C, Enables the Output Circuit and make the Buffer direction to (B <- A) which makes the PIN in Output Mode



AVR Microcontroller

In our course we will use Microcontroller AVR Atmega32. It has 32 DIO pins grouped as following:

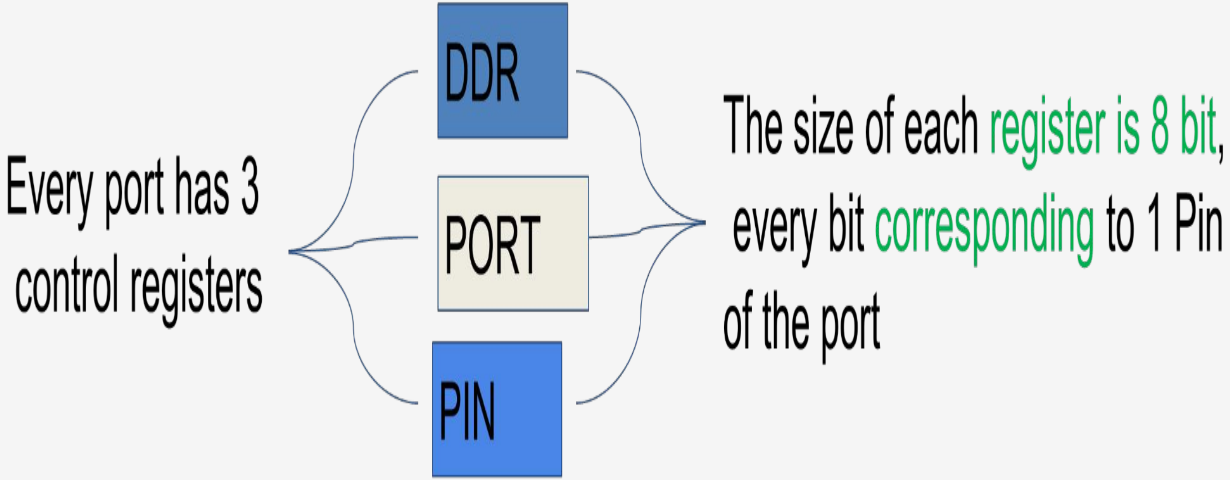
1- PORTA has 8 DIO Pins from A0 to A7

2- PORTB has 8 DIO Pins from B0 to B8

3- PORTC has 8 DIO Pins from C0 to C8

4- PORTD has 8 DIO Pins from D0 to D8

AVR DIO


Every port has 3 control registers
DDR
PORT
PIN
The size of each register is 8 bit, every bit corresponding to 1 Pin of the port

AVR DIO

1-DDR (Data Direction Register)

In this register we can define the pin is output or input.

Set 0 input

Set 1 output

2-PORT

This register is used in output mode to set the digital output value.

Set 0 This pin carry 0 V

Set 1 This pin carry 5 V

3-PIN

We use this register in case the pin is defined input.

If 0 The pin connected to 0 V

If 1 The pin connected to 5 V

AVR DIO

Example 1 :

Configuring Pin A0 Output 5V

DDRA=0b00000001;

PORTA=0b00000001;

Example 2 :

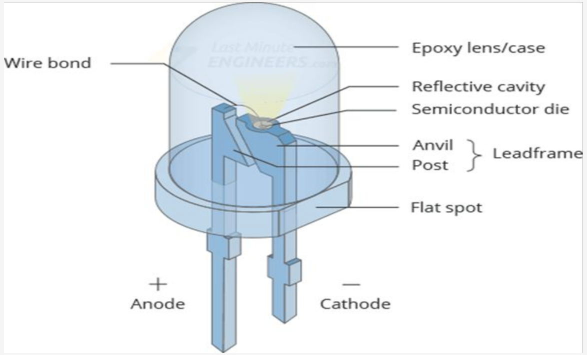
Configuring All PORTD Output, lower half connected to 5V and upper half connected to ground.

DDRA=255;   PORTA=0x0F;

Interfacing LEDs

LED Definition

Light Emitting Diode is an electrical element that emits light by supplying a voltage difference between its terminals.



Interfacing LEDs

DDRA = 0b00000001;

Configure pin A0 as output.

DDRA = 0b00000001;

Set A0 to digital 1 ie 5V.

Now If we are connecting the negative pin of led to ground and connecting positive pin of led to A0 , the led will be lighted up.

Then

DDRA = 0b00000001;

Set A0 to digital 0 i.e 0V.

Now the led will be closed up.

Super loop

Any C project in Embedded Systems application shall have an infinite loop called the super loop. This loop is a must even if you we will leave it empty ! This loop prevents the program counter (PC) from continues incrementing over the flash memory and execute a garbage code. i.e. the while(1) represents the end of the code.

```c
void main (void)
{
    /* Initialization Part */

    while (1)
    {
        /* Super Loop */
    }
}
```

Lab

Write a C code to turn on LED on Pin A0

AVR DIO

Using delay function

Software Technique the use a loop with effect just to halt the processor for certain time. We will use a library called “avr/delay.h” that provides two basic functions.

```
_delay_ms(_value_in_ms); /* Apply a delay in milli seconds */
_delay_us(_value_in_us); /* Apply a delay in micro seconds */
```

Note

Before using the delay library, we have to define our system frequency by writing this command:

```
#define F_CPU 12000000 /* Define a CPU frequency of 12 Mega Hertz */
```

Lab

Write a C code to turn on LED on Pin A0 for 1 second and then turn it off

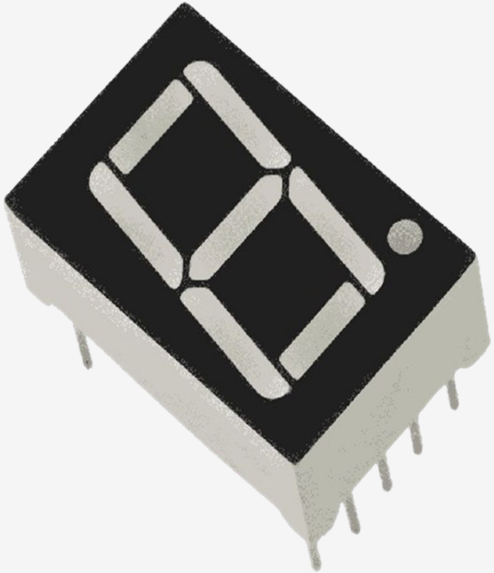
Lab

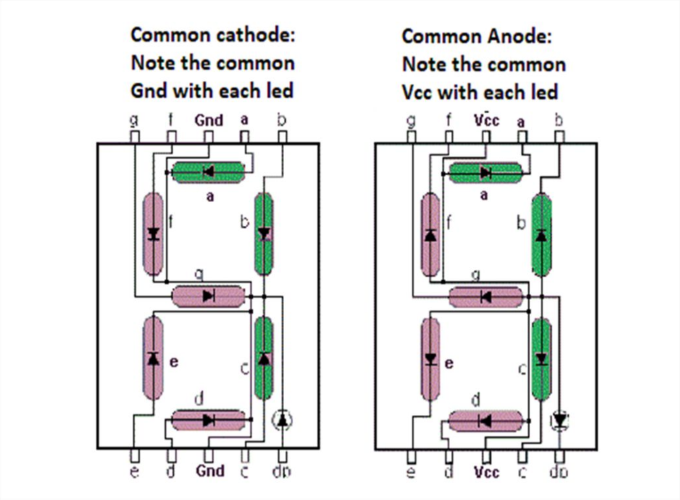
- Write a C code to blink a LED Every 1 second

Lab

Write a C Code that apply Some LED animations

Interfacing 7 segment

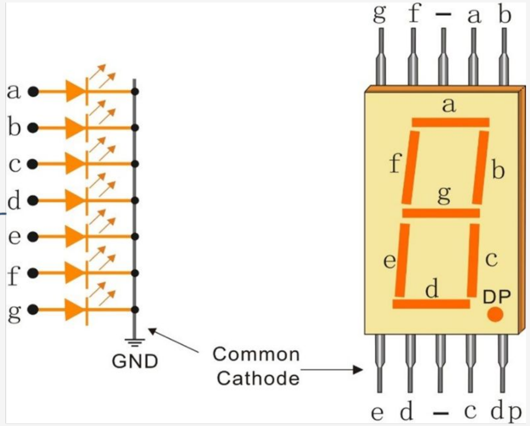




Interfacing 7 segment

7–segment is common cathode :

Assuming Connecting the 7- Segment lines (a to g) to PD0 to PD6 in the microcontroller kit



7 Segment Truth Table

s	BCD	G	F	E	D	C	B	A
0	0000	0	1	1	1	1	1	1
1	0001	0	0	0	0	1	1	0
2	0010	1	0	1	1	0	1	1
3	0011	1	0	0	1	1	1	1
4	0100	1	1	0	0	1	1	0
5	0101	1	1	0	1	1	0	1
6	0110	1	1	1	1	1	0	1
7	0111	0	0	0	0	1	1	1
8	1000	1	1	1	1	1	1	1
9	1001	1	1	0	1	1	1	1

Lab

write a code to display on 7-segment numbers from 0 to 9 with delay 1 second before changing number

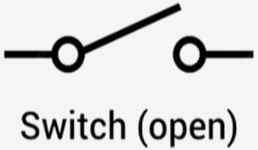
Mechanical Switch

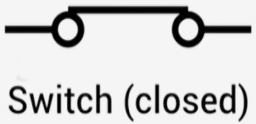
Mechanical Switch is an electrical component that can connect or break an electrical circuit.

Open

Switch state

Close





Mechanical Switch

Tactile switch

Mechanical Switch

Push Button

Mechanical Switch

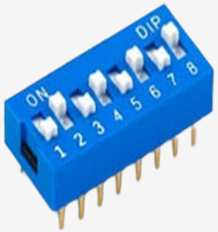
Paddle switch

Mechanical Switch

Rocker switch

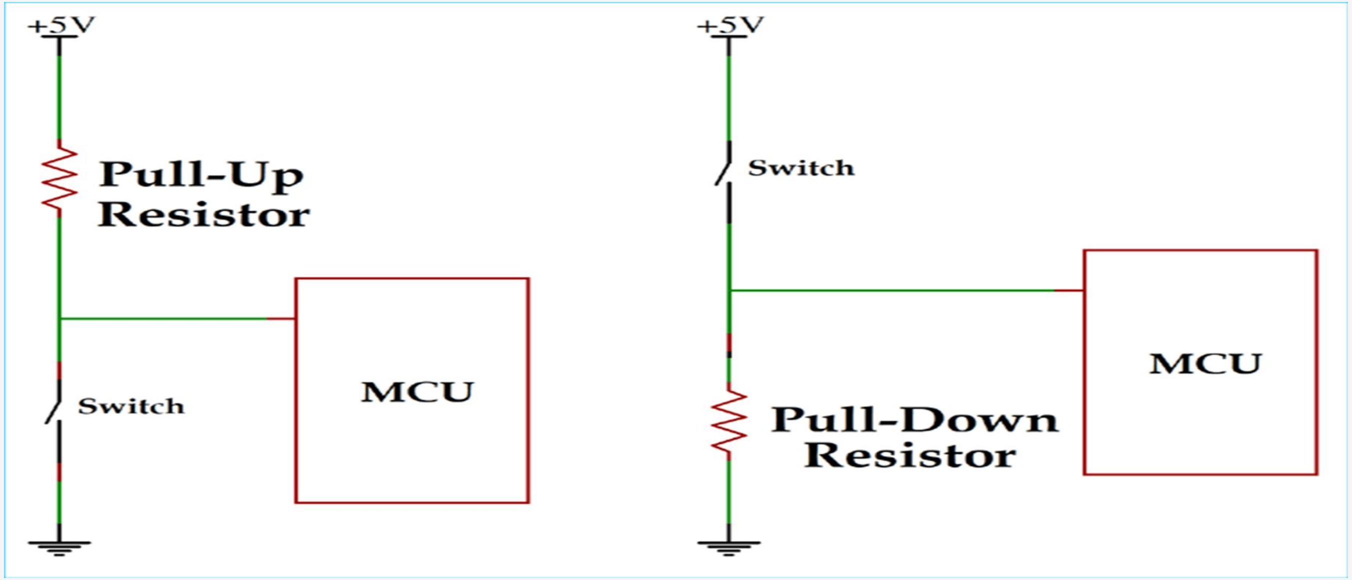
Mechanical Switch

DIP switch



Interfacing Mechanical Switch

Switch shall be connected by pull up or pull down resistor to avoid short circuits.



Interfacing Mechanical Switch

In AVR Microcontroller, all DIO pins have internal pull up resistors that can be activated or not.

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Input Digital Pin

Floating

DDR -> 0

Configure PIN as input.

Internal pull up

DDR -> 0
PORT -> 1

Activate internal pull up.

Interfacing Mechanical Switch

Reading Input PIN

The registers PINA, PINB, PINC and PIND are used to check the status of the input pins. If the corresponding bit for a certain pin is 0, then the pin is connected to GND. If the corresponding bit for a certain pin is 1, then the pin is connected to VCC.

```
/* Check if Pin A0 is connected to GND */
if ( (PINA & 0b00000001) == 0 )

/* Check if Pin B3 is connected to VCC */
if ( (PINB & 0b00001000) == 0 )
```

Lab

Write a code that uses a DIP switch to control a string of 8 LEDs. When the DIP switch is On the LED string shall be flashing every 500 ms. When the DIP switch off the LED string shall be also off.

Thank You

Do you have any questions?

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